LEARNING EQUIVARIANT NON-LOCAL ELECTRON DENSITY FUNCTIONALS

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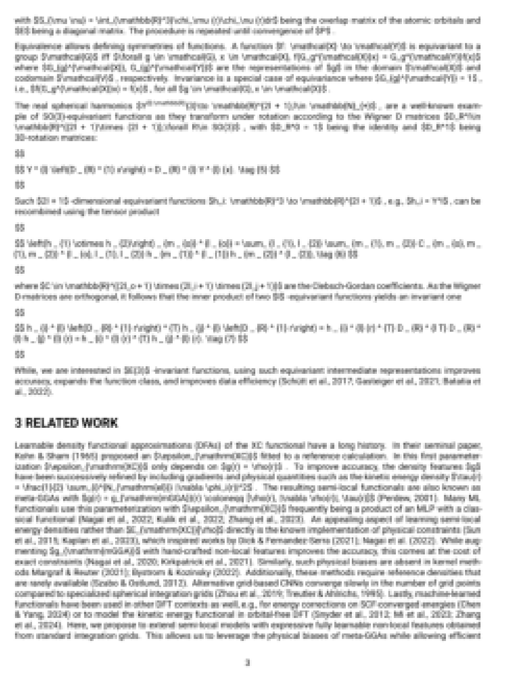
Paper under double-blind review

ABSTRACT

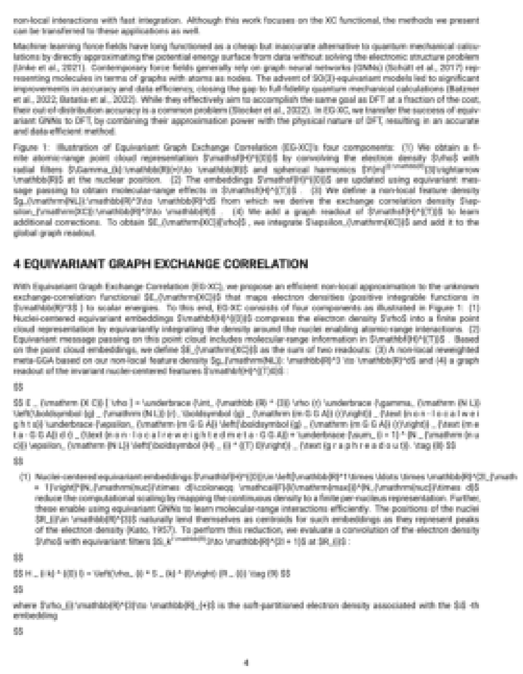
The accuracy of density functional theory hinges on the approximation of non-local contributions to the exchange-correlation (XC) functional. To date, machine-learned and human-designed approximations suffer from insufficient accuracy, limited scalability, or dependence on costly reference data. To address these issues, we introduce Equivariant Graph Exchange Correlation (EG-XC), a novel non-local XC functional based on equivariant graph neural networks (GNNs). Where previous works relied on semi-local functionals or fixed-size descriptors of the density, we compress the electron density into an SO(3)-equivariant nuclei-centered point cloud representation for efficient non-local atomic-range interactions. By applying an equivariant GNN on this point cloud, we capture molecular-range interactions in a scalable and accurate manner. To train EG-XC, we differentiate through a self-consistent field solver requiring only energy targets. In our empirical evaluation, we find EG-XC to accurately reconstruct 'gold-standard' CCSD(T) energies on MD17. On out-of-distribution conformations of 3BPA, EG-XC reduces the relative MAE by 35% to 50% . Remarkably, EG-XC excels in data efficiency and molecular size extrapolation on QM9, matching force fields trained on 5 times more and larger molecules. On identical training sets, EG-XC yields on average 51% lower MAEs.

1 INTRODUCTION

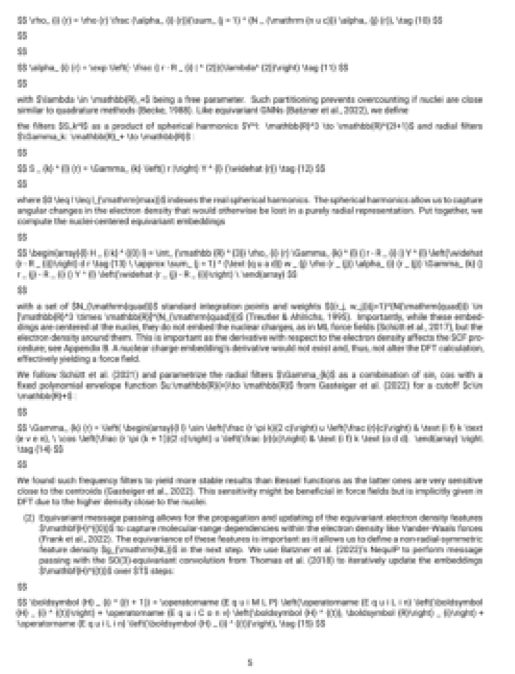
2 BACKGROUND

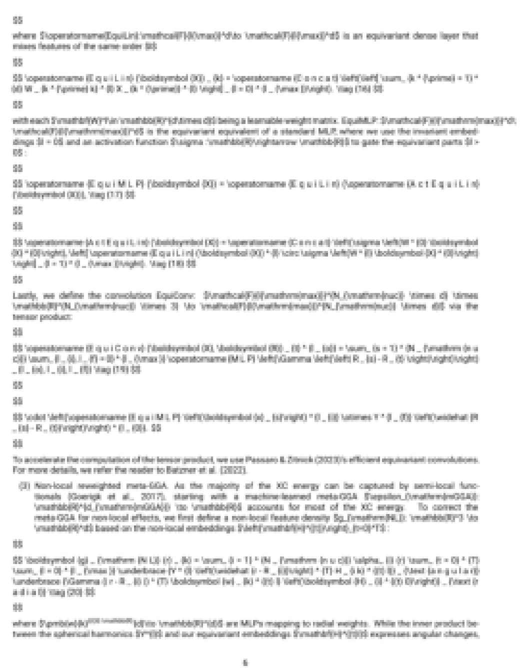


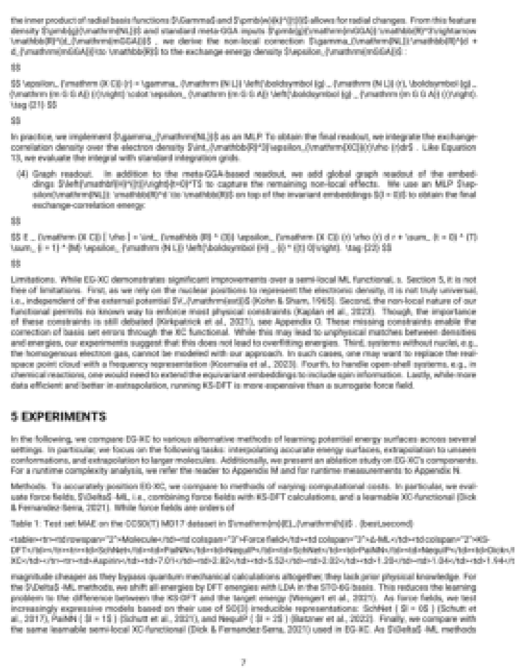
3 RELATED WORK



4 EQUIVARIANT GRAPH EXCHANGE CORRELATION







5 EXPERIMENTS

6 DISCUSSION